

Lecture 7

Thursday 7/16/26

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US Particle Accelerator School
Accelerator Physics – Summer 2026

Dispersion vector form

$$\bar{d} = \begin{cases} \begin{pmatrix} 0 \\ 0 \end{pmatrix} & \text{for drift space and quadrupole} \\ \begin{pmatrix} \frac{1}{\rho K_x} (1 - \cos \sqrt{K_x} s) \\ \frac{1}{\rho \sqrt{K_x}} \sin \sqrt{K_x} s \end{pmatrix} & \text{for dipole with } K_x > 0, \\ \begin{pmatrix} \frac{1}{\rho |K_x|} (-1 + \cosh \sqrt{|K_x|} s) \\ \frac{1}{\rho \sqrt{|K_x|}} \sinh \sqrt{|K_x|} s \end{pmatrix} & \text{for dipole with } K_x < 0. \end{cases}$$

(SY Lee Sec. 2.IV.1)

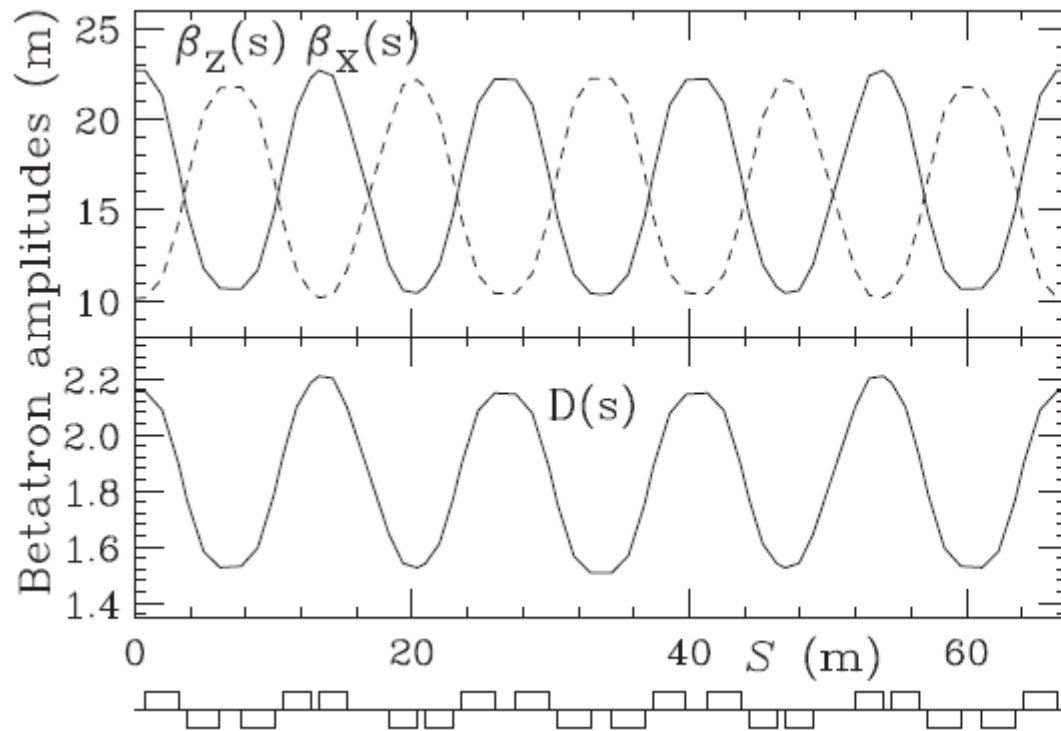
Dispersion through FODO

Previously, sector bend and drift are equivalent. With dispersion:

$$M = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2f} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & L_1 & \frac{1}{2}L_1\theta \\ 0 & 1 & \theta \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{f} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & L_1 & \frac{1}{2}L_1\theta \\ 0 & 1 & \theta \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2f} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(SY Lee)

Figure 2.5: The betatron amplitude functions for one superperiod of the AGS lattice, which made of 20 combined-function magnets. The upper plot shows β_x (solid line) and β_z (dashed line). The middle plot shows the dispersion function D_x . The lower plot shows schematically the placement of combined-function magnets. Note that the superperiod can be well approximated by five regular FODO cells. The phase advance of each FODO cell is about 52.8° .



Dispersion with linear magnet errors

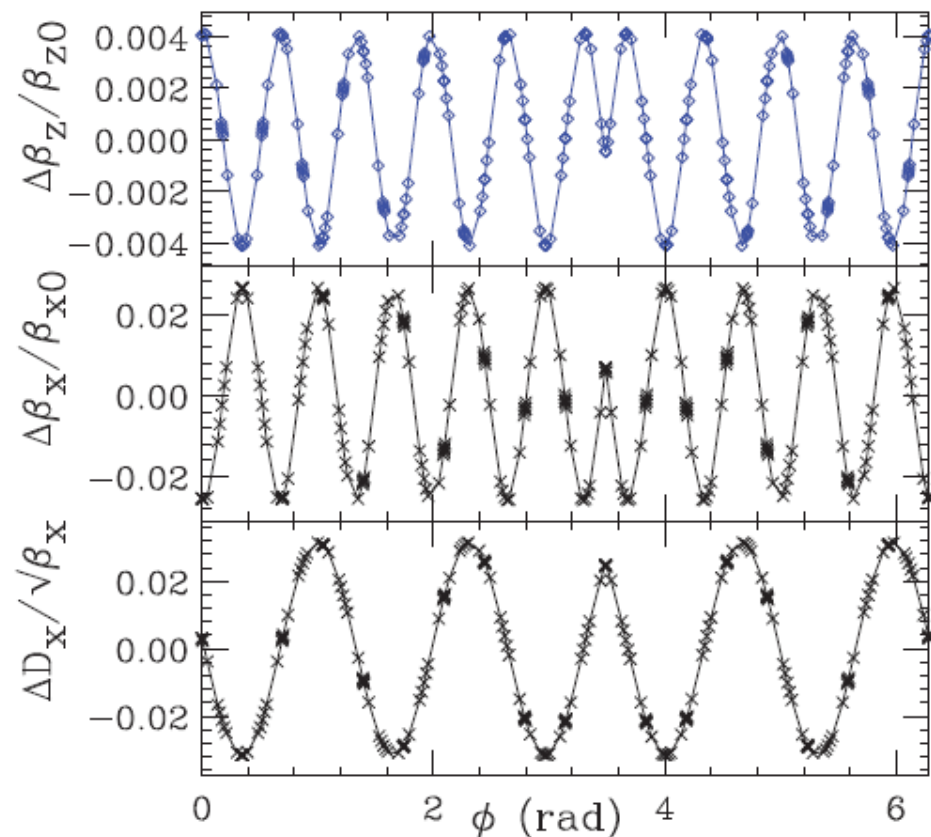
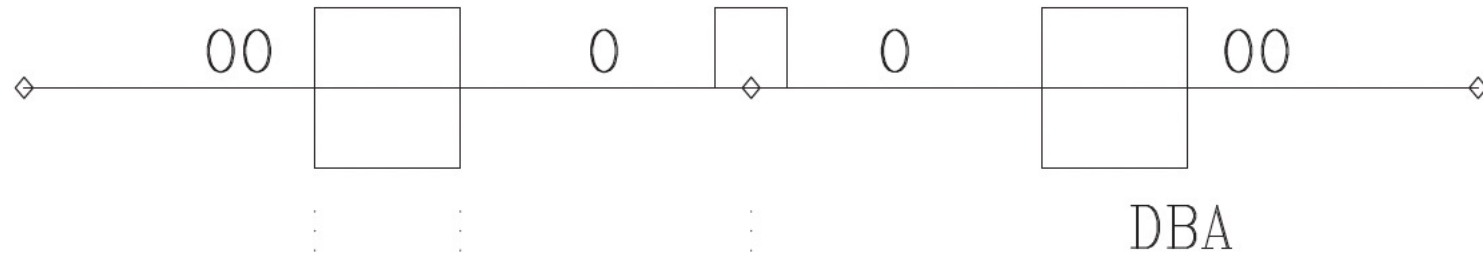


Figure 2.20: Perturbation of betatron amplitude functions vs ϕ (either $\phi_x = \frac{1}{\nu_x} \int_0^s ds/\beta_x$ or $\phi_z = \frac{1}{\nu_z} \int_0^s ds/\beta_z$) resulting from 1% decrease in gradient strength of the 10th focusing quadrupole. The betatron amplitude function perturbation is dominated by harmonics nearest $[2\nu_x]$ and $[2\nu_z]$. Since $\beta_x/\beta_z \sim 6.37$ at the focusing quadrupole location, the resulting error $\Delta\beta_x/\beta_x$ is about $6.37\Delta\beta_z/\beta_z$. A single kick at the error quadrupole location can be identified in the top 2 plots. The bottom plot shows the effect of quadrupole error on dispersion function shown as $\Delta D_x/\sqrt{\beta_x}$ vs $\phi = \phi_x$. A single kick at the error quadrupole location is visible to the dispersion closed orbit.

SY Lee Fig. 2.20

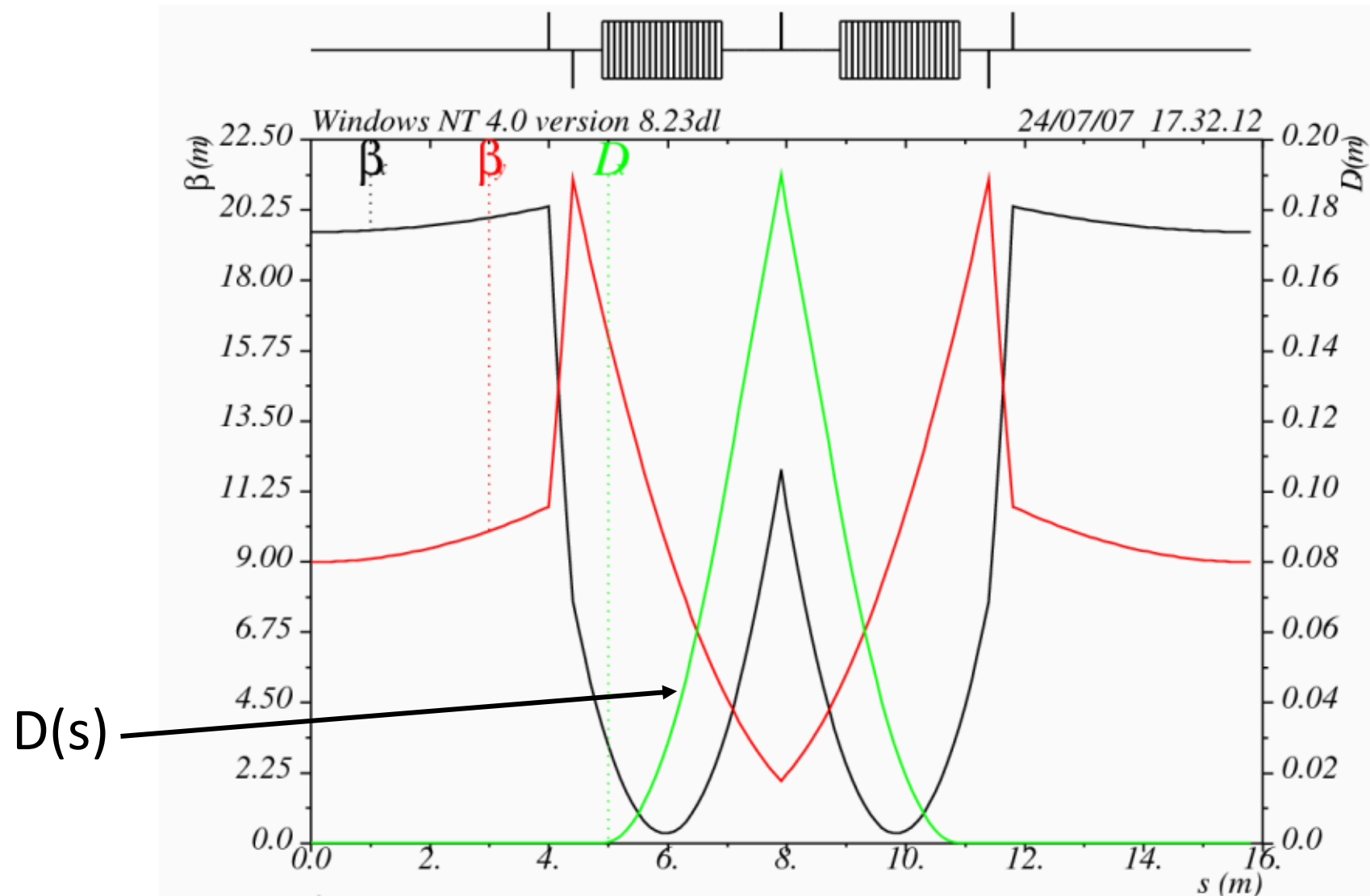
DBA with single quad



$$\begin{pmatrix} D_c \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1/(2f) & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & L_1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & L & L\theta/2 \\ 0 & 1 & \theta \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

DBA with single quad

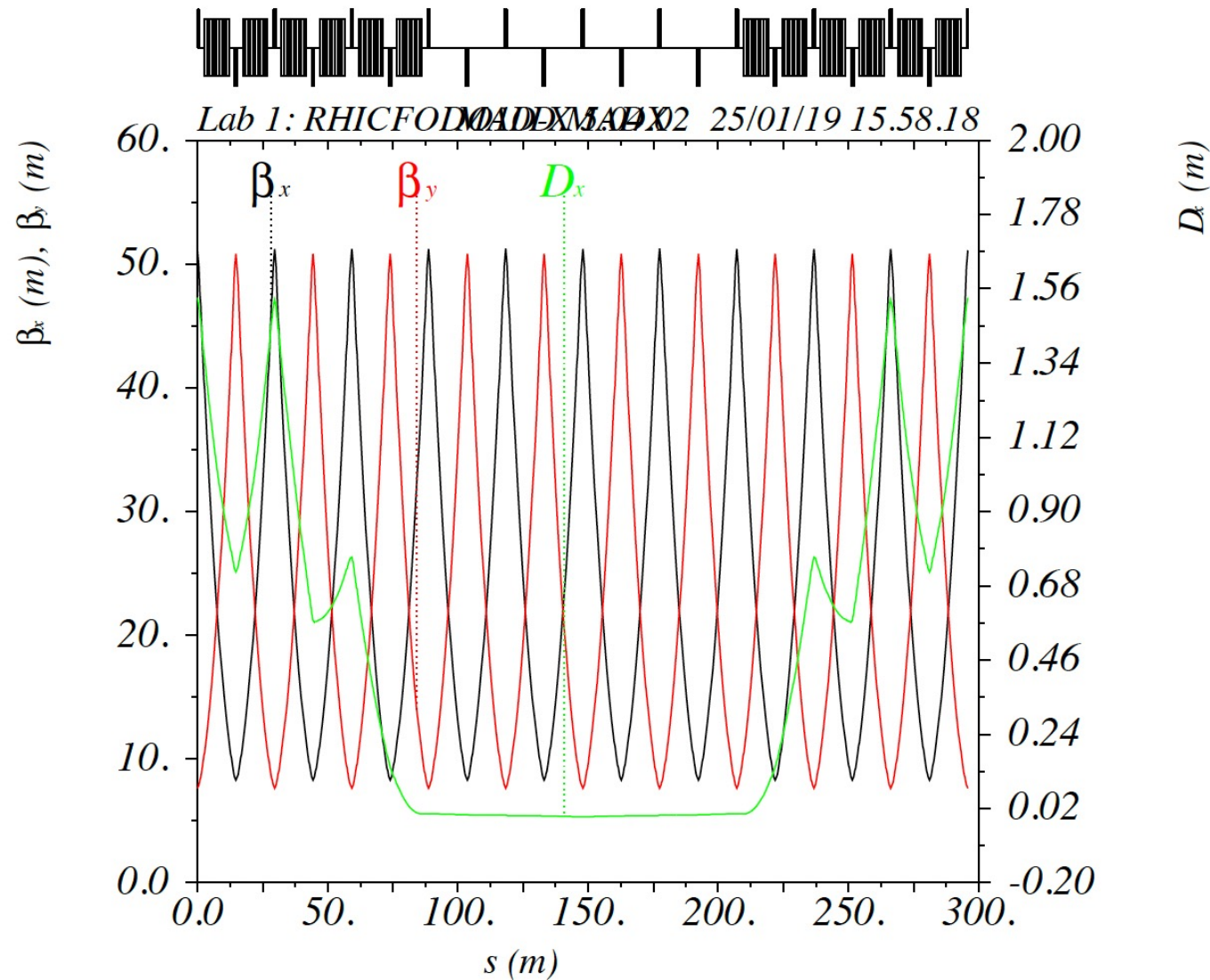
<https://uspas.fnal.gov/materials/12MSU/DampingRings-Lecture2.pdf>



Dispersion suppression

2019 USPAS course with MADx lab exercises, plus lots of good visualizations of tune, resonances, etc.

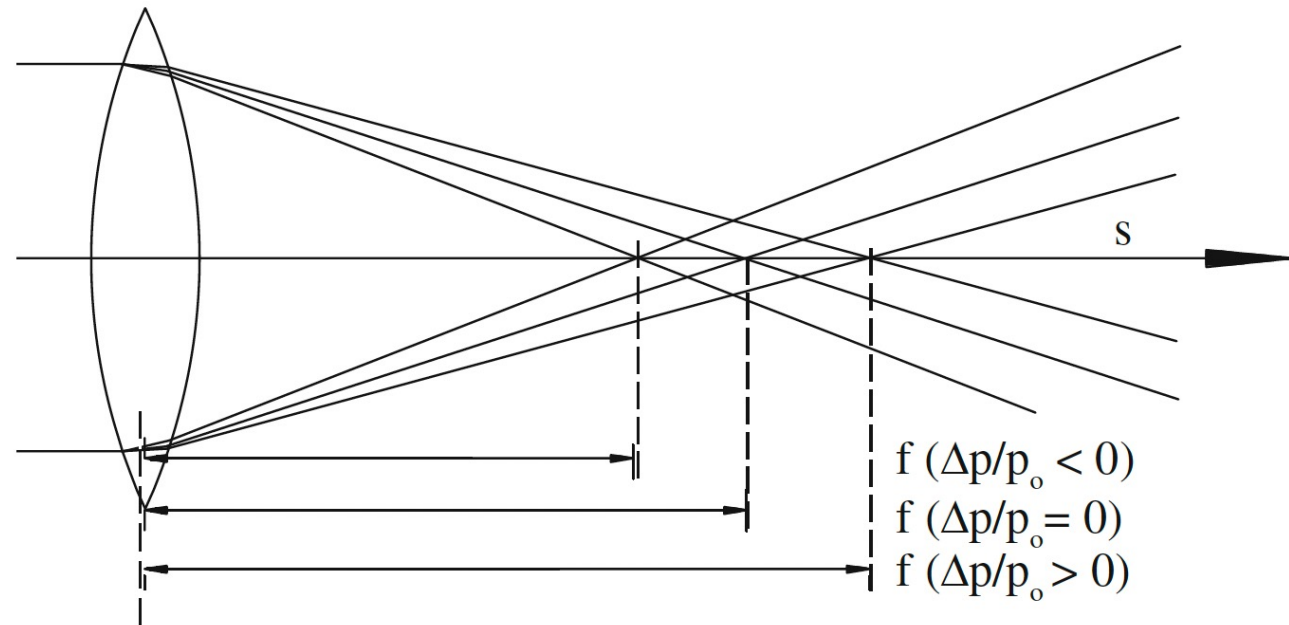
<https://toddsatogata.net/2019-USPAS/>



Chromatic aberration

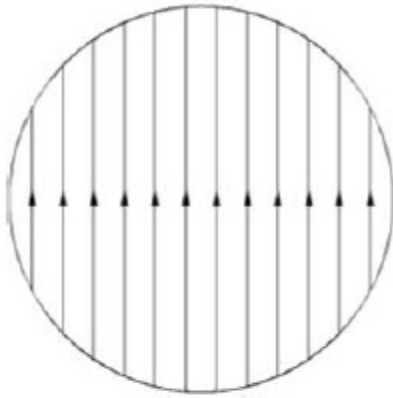
Wiedemann

Fig. 15.8 Chromatic focusing errors

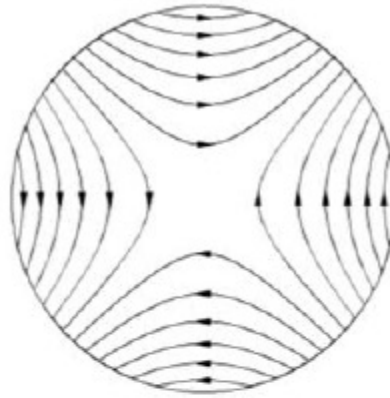


Multipole fields

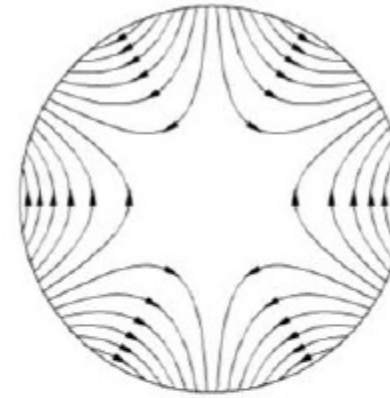
B_1 : normal dipole



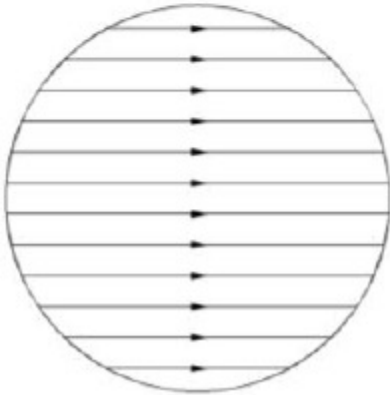
B_2 : normal quadrupole



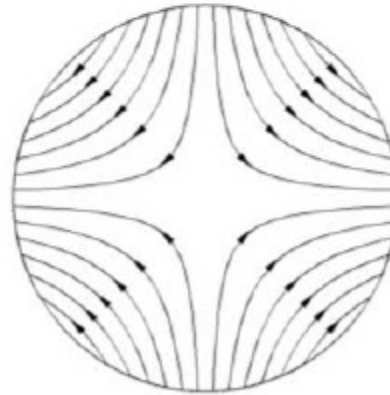
B_3 : normal sextupole



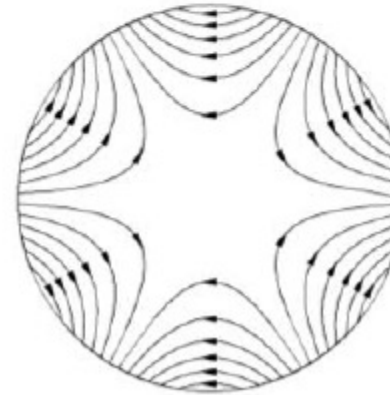
A_1 : skew dipole



A_2 : skew quadrupole



A_3 : skew sextupole



Chromaticity correction with sextupoles

Wiedemann

Fig. 15.8 Chromatic focusing errors

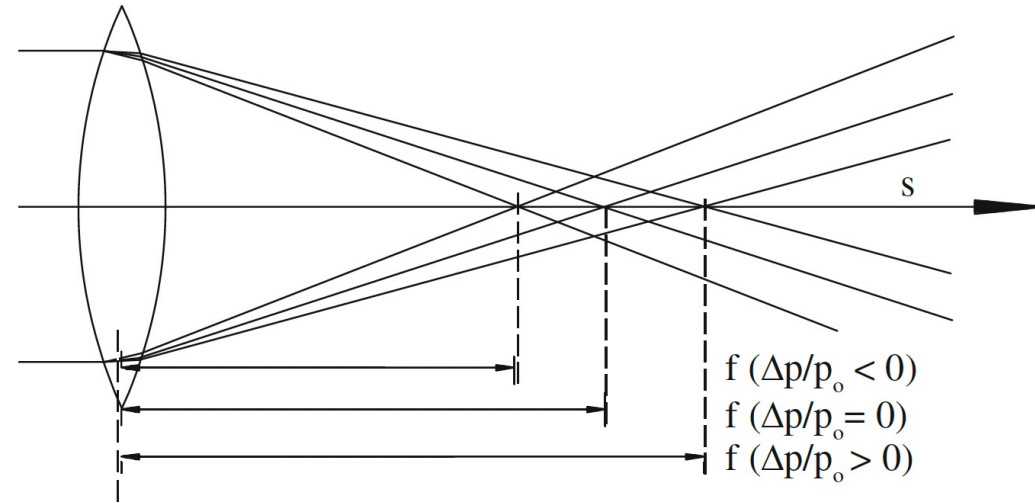
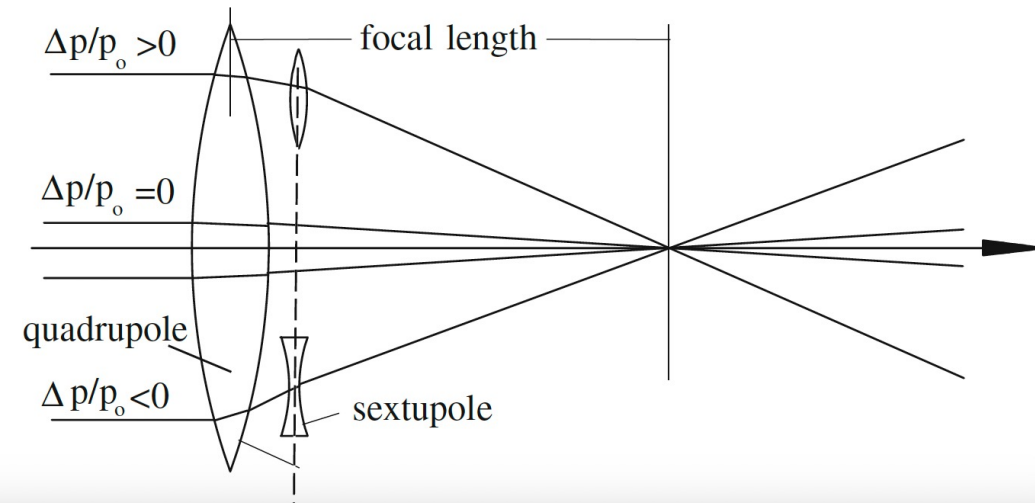
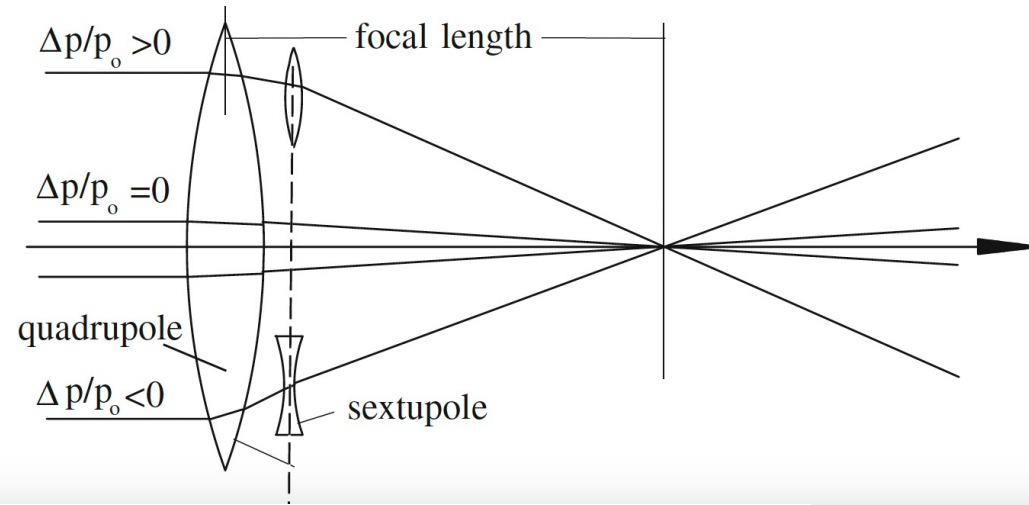


Fig. 15.9 Chromaticity correction with sextupoles



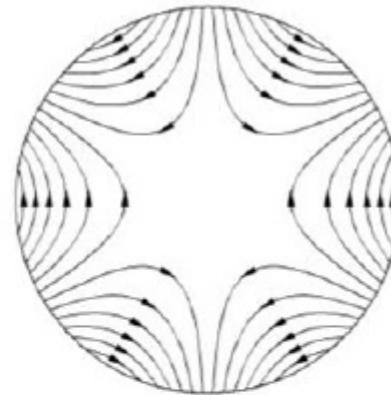
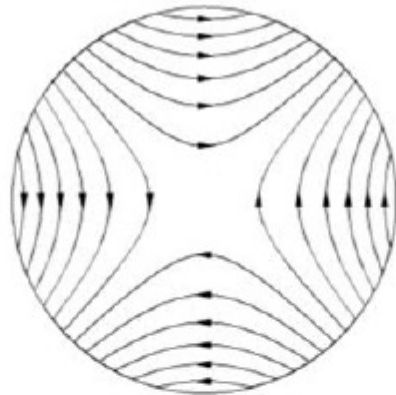
Chromaticity correction with sextupoles

Fig. 15.9 Chromaticity correction with sextupoles



B_2 : normal quadrupole

B_3 : normal sextupole



Look at field
direction at
midplane

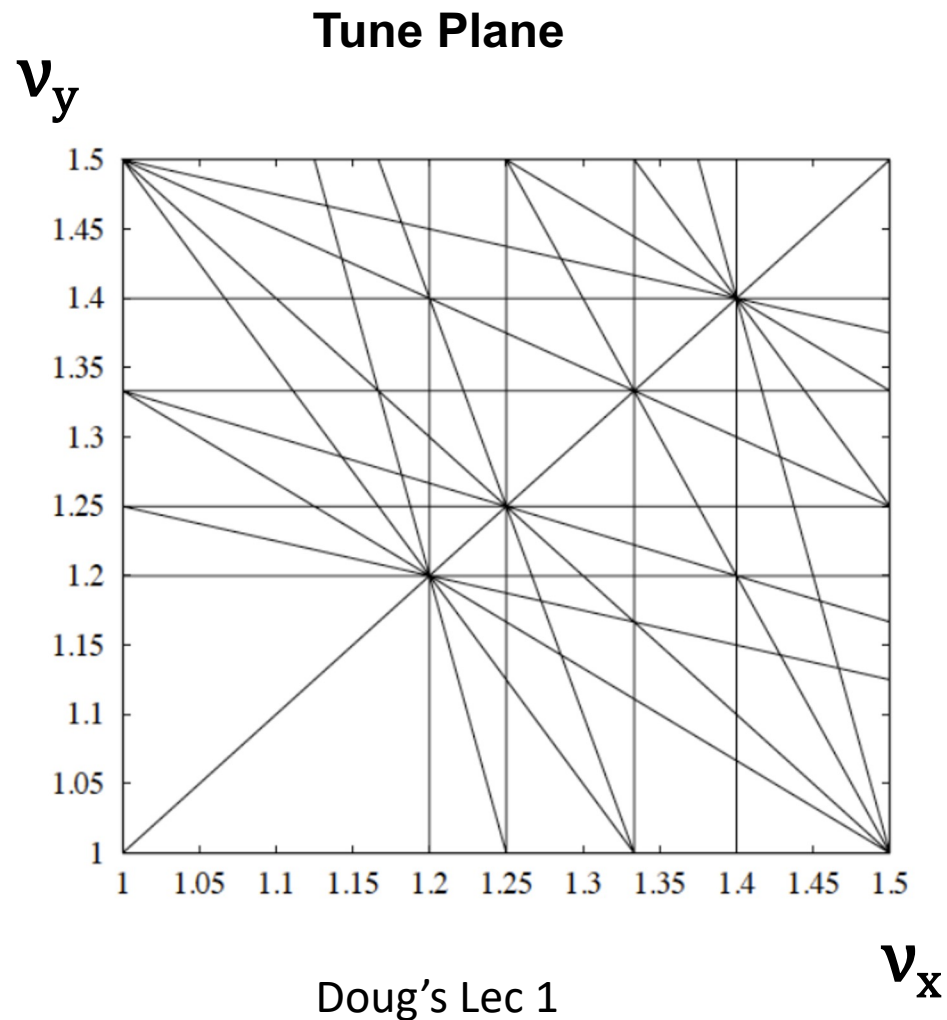
Non-linear resonances from sextupoles

SY Lee, ℓ is an integer

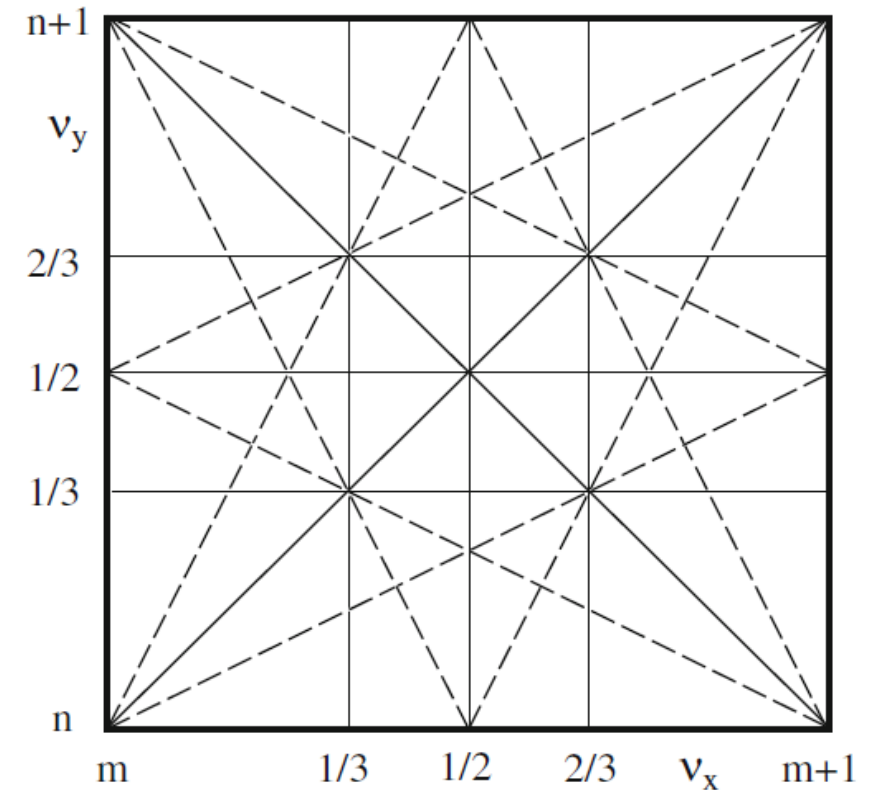
Table 2.3: Resonances due to sextupoles and their driving terms

Resonance	Driving term	Lattice	Amplitude	Classification
$\nu_x + 2\nu_z = \ell$	$\cos(\Phi_x + 2\Phi_z)$	$\beta_x^{1/2} \beta_z$	$J_x^{1/2} J_z$	sum resonance
$\nu_x - 2\nu_z = \ell$	$\cos(\Phi_x - 2\Phi_z)$	$\beta_x^{1/2} \beta_z$	$J_x^{1/2} J_z$	difference resonance
$\nu_x = \ell$	$\cos \Phi_x$	$\beta_x^{1/2} \beta_z; \beta_x^{3/2}$	$J_x^{1/2} J_z, J_x^{3/2}$	parametric resonance
$3\nu_x = \ell$	$\cos 3\Phi_x$	$\beta_x^{3/2}$	$J_x^{3/2}$	parametric resonance

Non-linear resonances from sextupoles

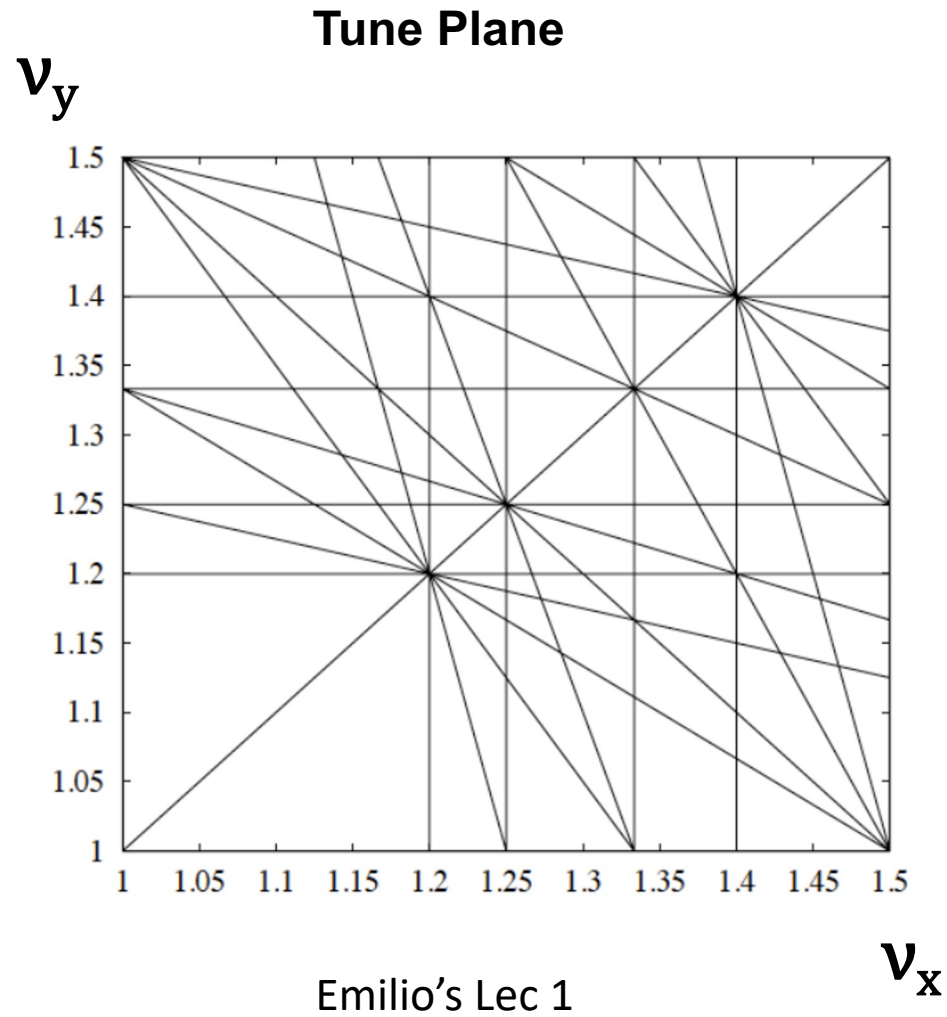


Spot our new resonances



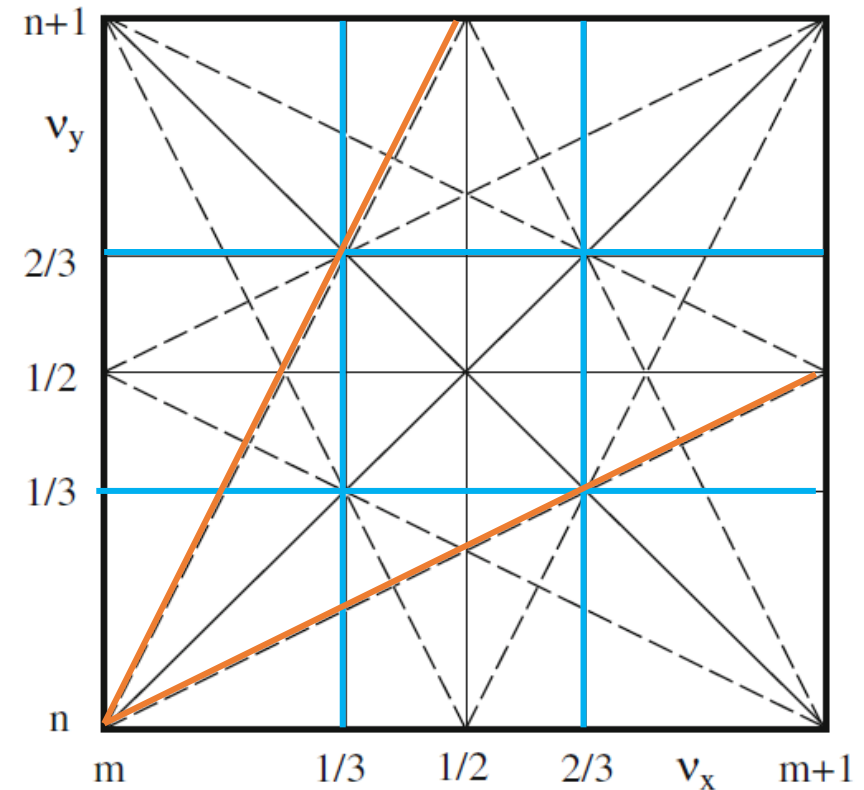
Wiedemann book Fig. 16.2

Non-linear resonances from sextupoles



$$v_x \pm 2v_y = \ell$$

$$3v_x = \ell, 3v_y = \ell$$



Wiedemann book Fig. 16.2

Building magnets

- Need to consider cases besides vacuum:

$\mu = \mu_r \mu_0$	material-dependent magnetic permeability	[H/m]
μ_r	relative permeability	[unitless]
$\mathbf{H}$	H field/magnetic field	[A/m]
$\mathbf{B} = \mu \mathbf{H}$	magnetic flux density/magnetic field	[T] or [G]

 This is the magnetic field we care about (for building magnets)

Maxwell's equations

$$\nabla \cdot \mathbf{D} = \rho$$

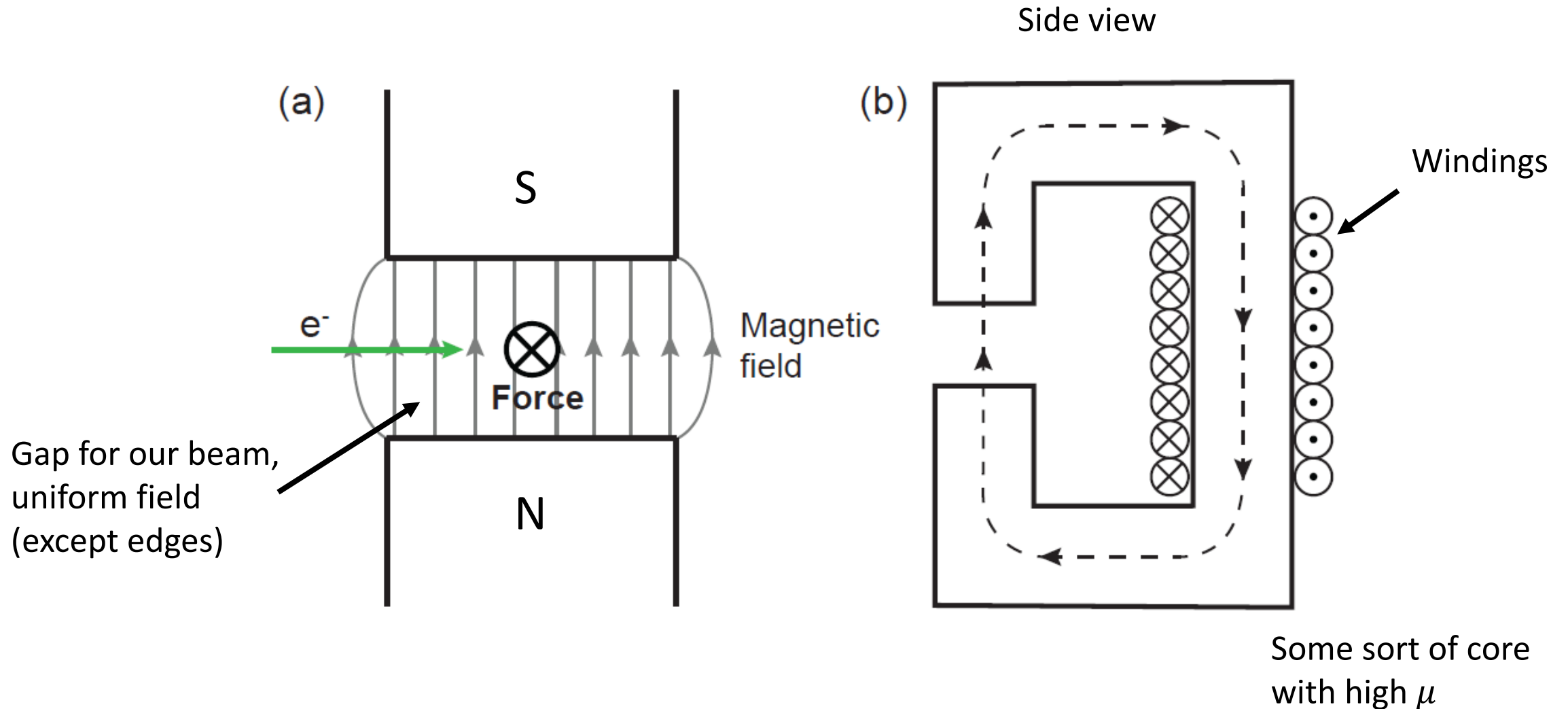
$$\nabla \times \mathbf{H} = i\omega \mathbf{D} + \mathbf{J}$$

$$\nabla \times \mathbf{E} = -i\omega \mathbf{B}$$

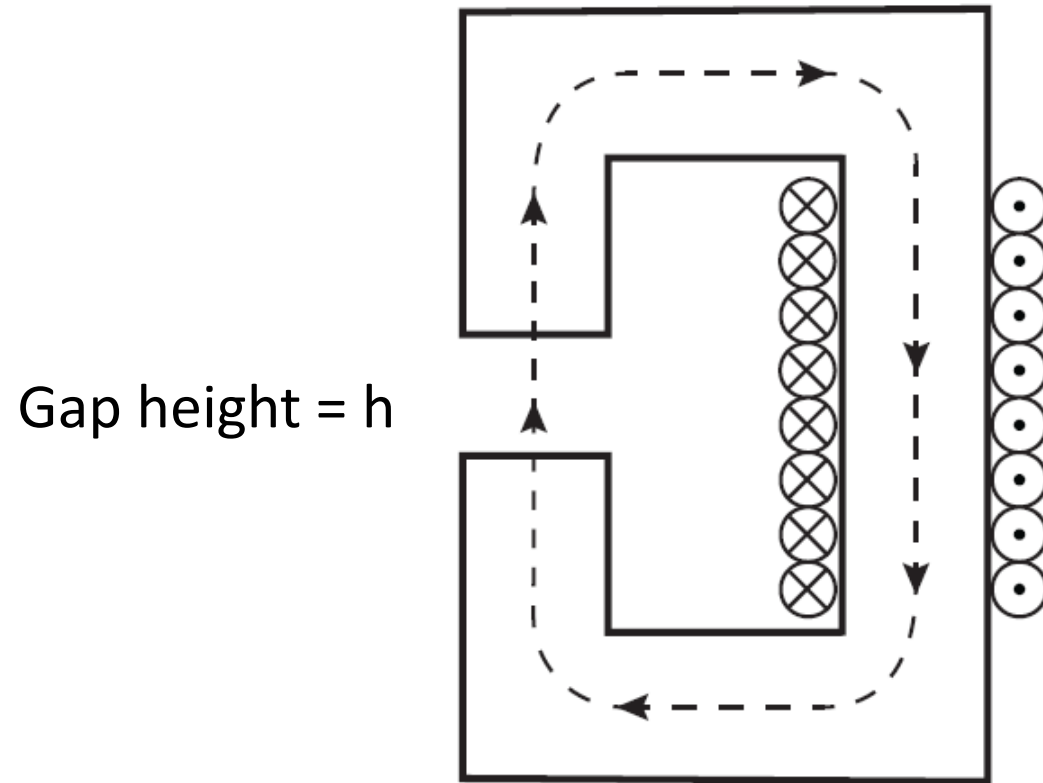
$$\nabla \cdot \mathbf{B} = 0$$

 For building the magnet itself, need some source

Dipole magnets – simple view



Field in the gap from Amperian loop



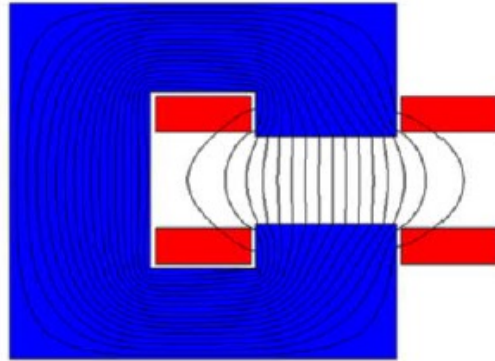
Some N turns, $I_{enc} = NI$

$$\text{Get } B = \mu_0 \frac{NI}{h}$$

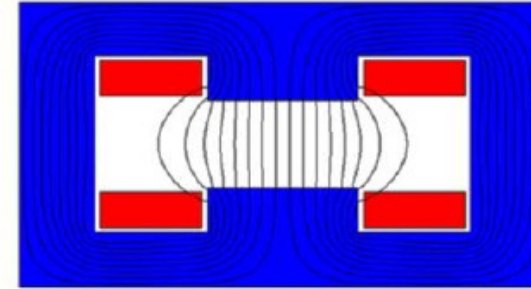
Some dipole configurations

Looking along beam path s

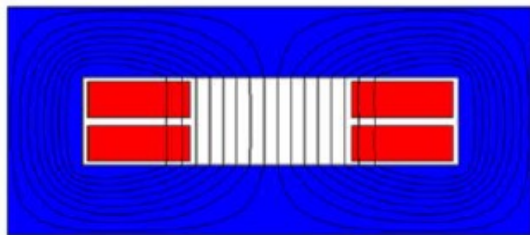
Basic design: flat pole faces



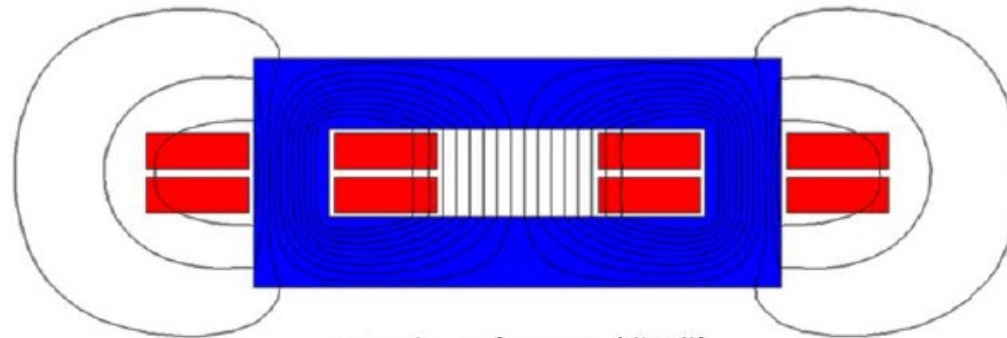
"C"



"H"



window frame
("O")

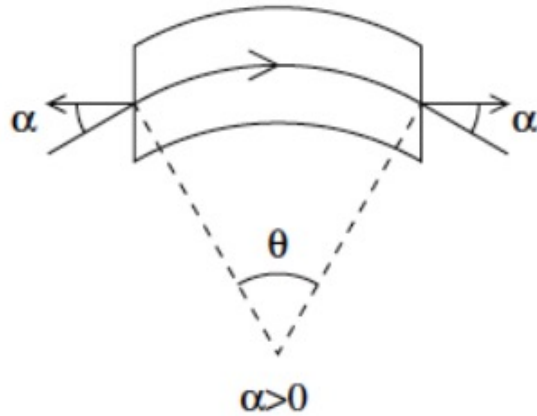


window frame ("O")
with windings on both backlegs

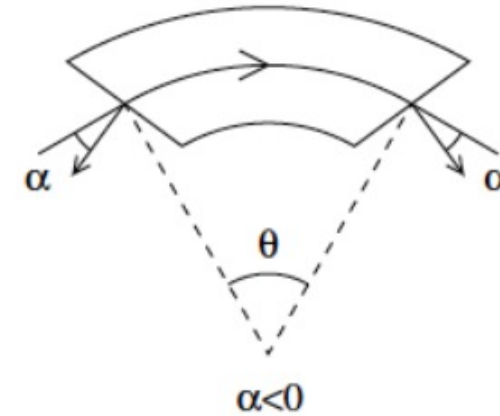
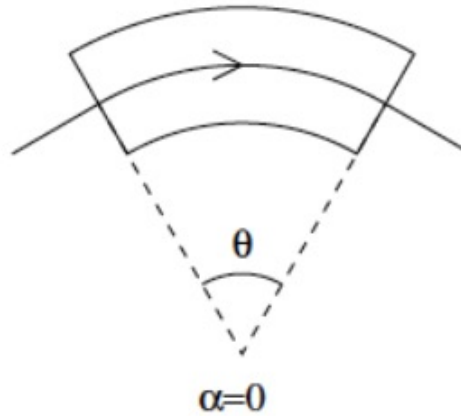
Figure: https://indico.cern.ch/event/1101643/contributions/4635257/attachments/2372240/4058445/JAI_course_Jan_2022_handouts.pdf

Dipole edge effects

“Rectangular” dipole



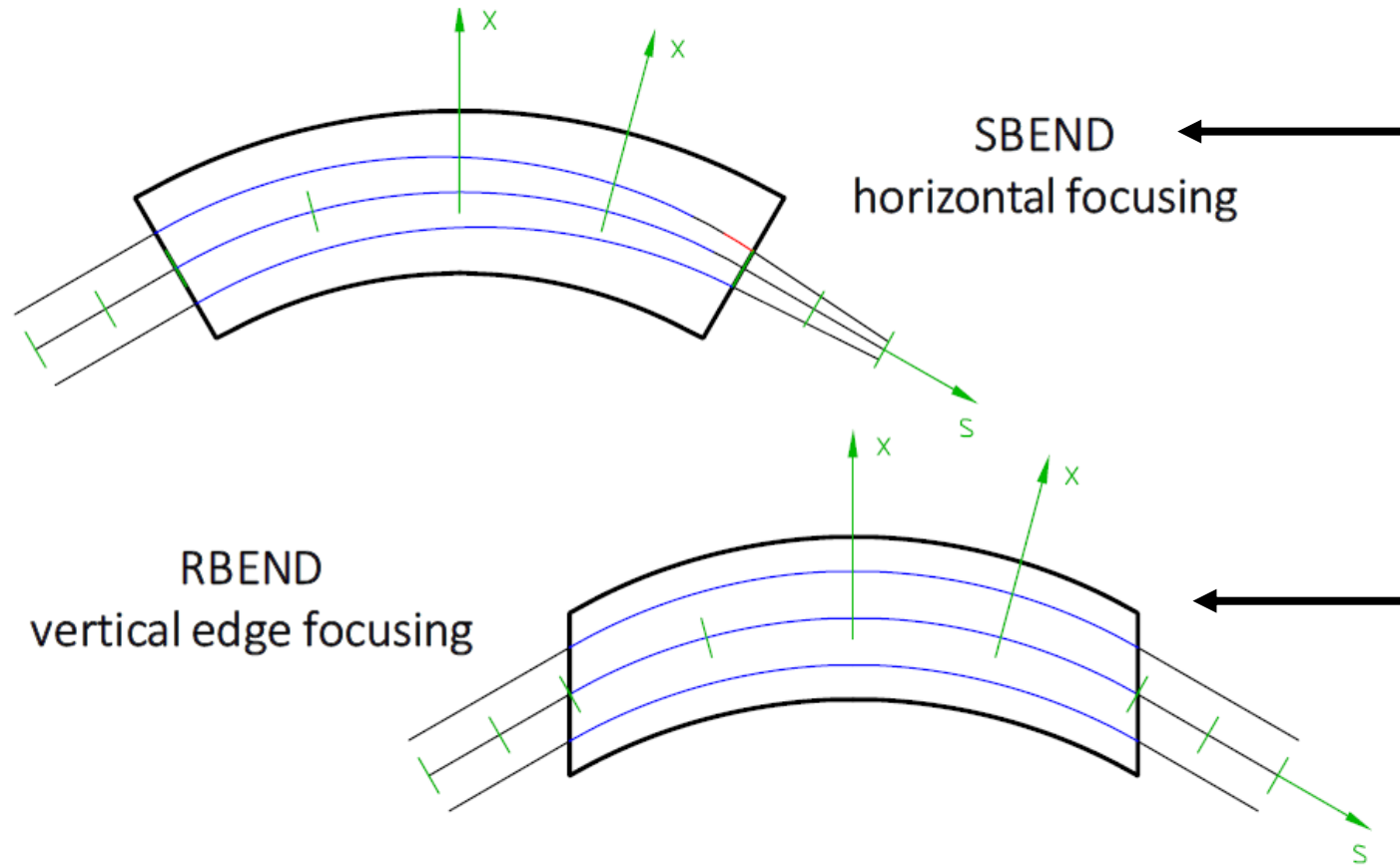
Sector dipole



Different effects depending on angle of faces relative to ideal beam path

Figure: https://www.toddsatogata.net/2019-USPAS/Lectures/Lecture_6_23_PM_satogata.pdf

Dipole edge effects



Path-length based, as we discussed in Lecture 2 (betatron) and Lecture 3

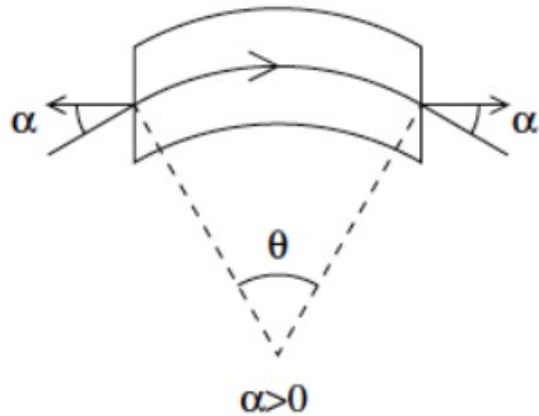
Also like the betatron, but this time entering/exiting the magnet rather than traveling in a circle between angled pole pieces

- and anything in between (playing with the edge angles) -

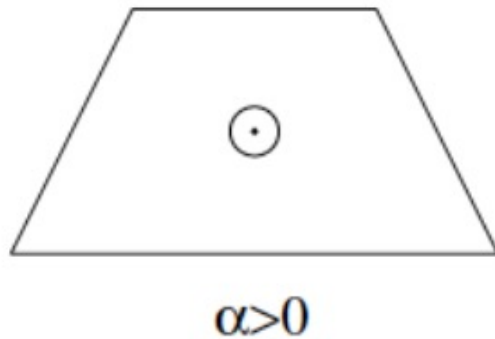
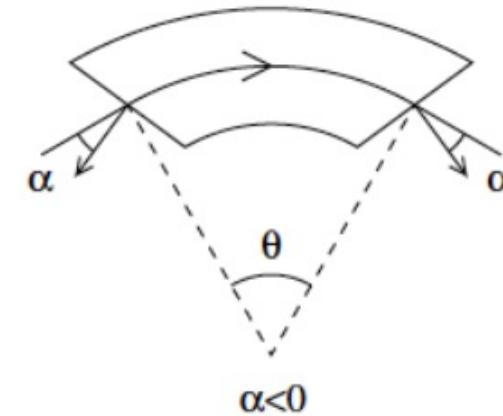
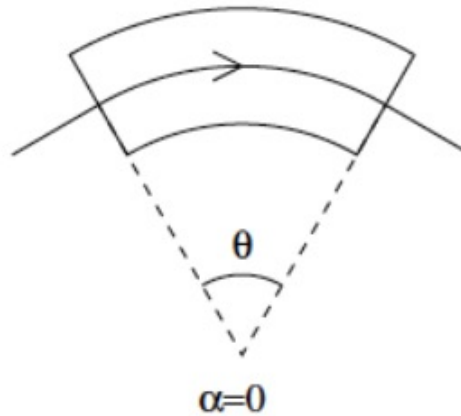
Figure: https://indico.cern.ch/event/1101643/contributions/4635257/attachments/2372240/4058445/JAI_course_Jan_2022_handouts.pdf

Dipole edge effects

“Rectangular” dipole



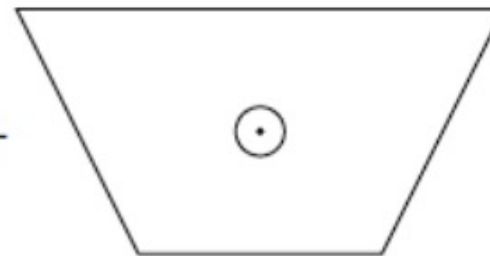
Sector dipole



=



+



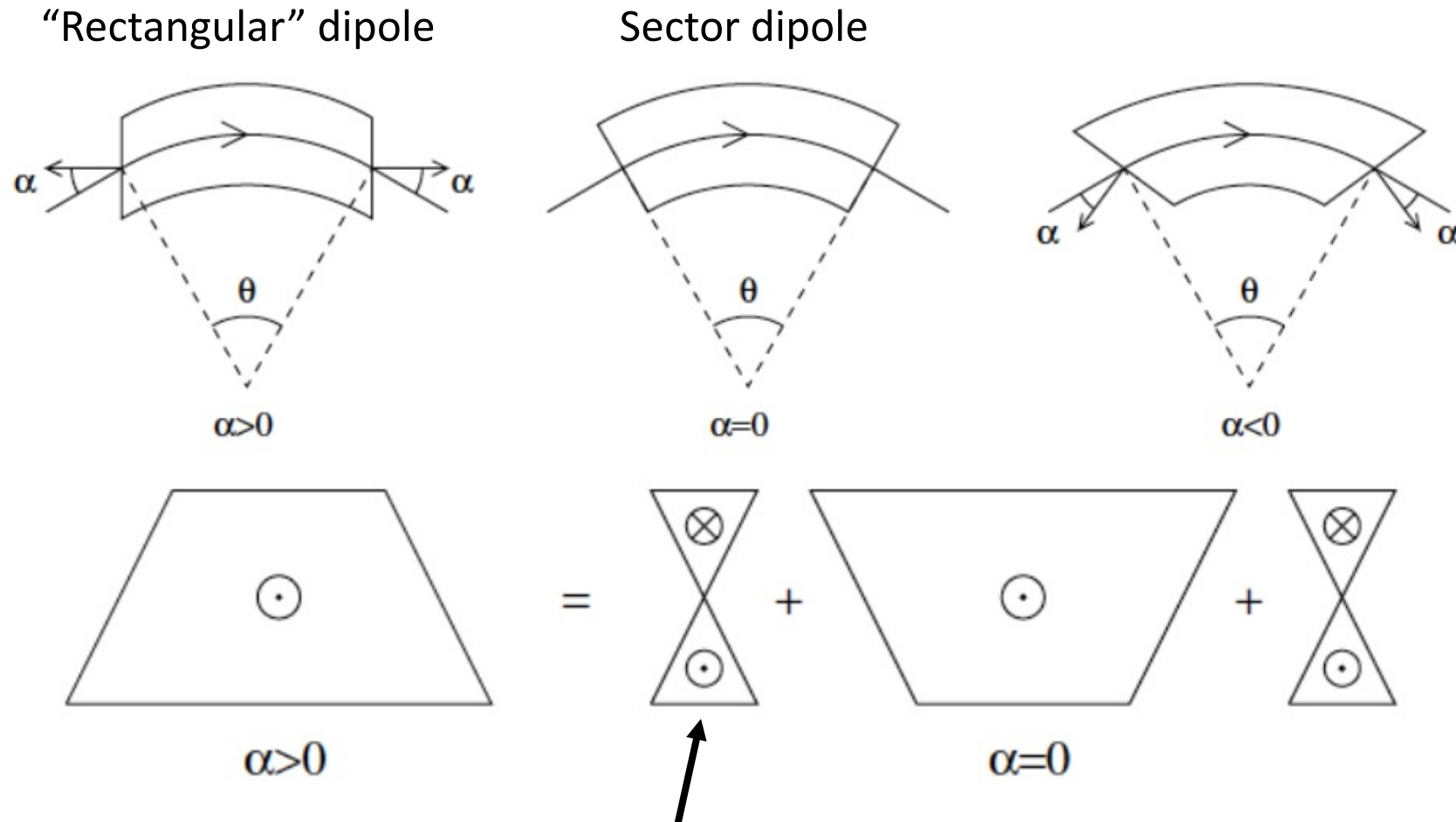
+



Represent non-sector dipole as sector dipole plus wedges for ends (superposition)

Figure: https://www.toddsatogata.net/2019-USPAS/Lectures/Lecture_6_23_PM_satogata.pdf

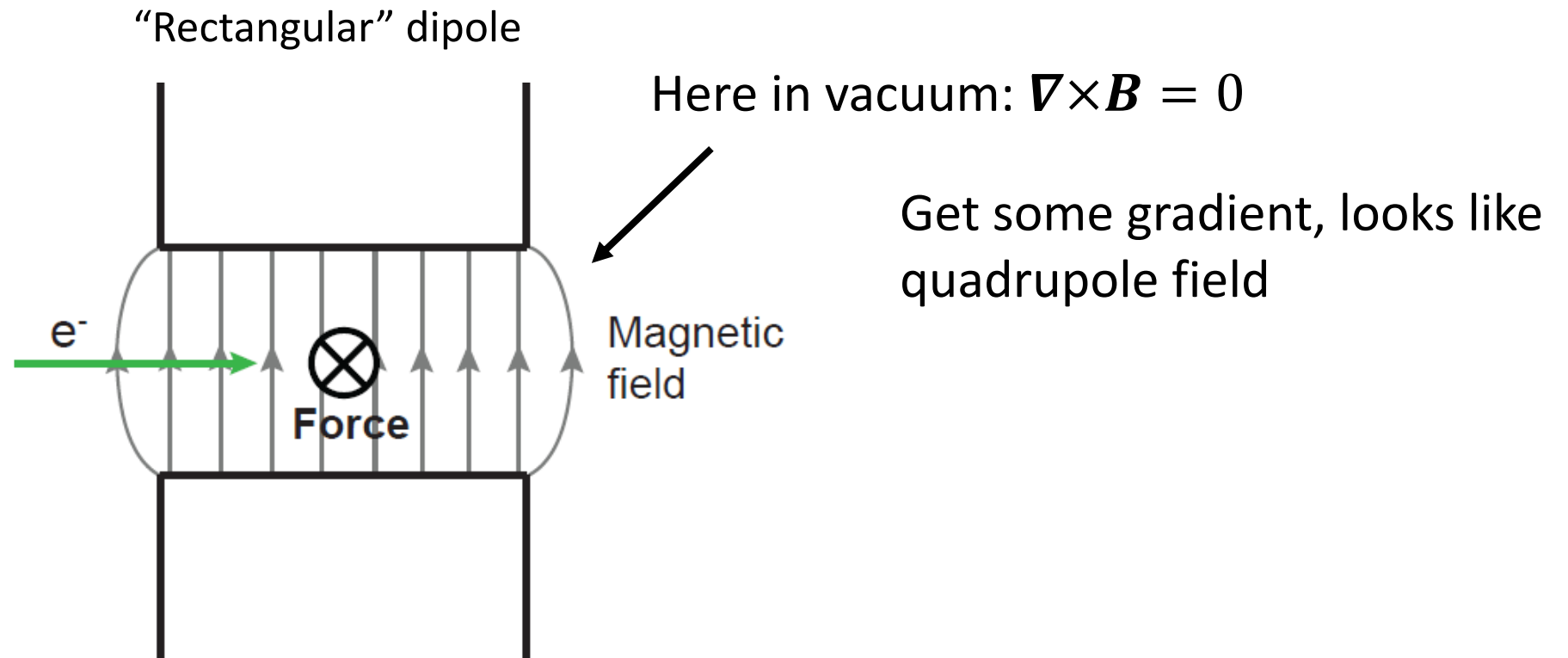
Dipole edge effects



Thin wedge: some kick from gradient like our quad misalignment discussion in Lec 5

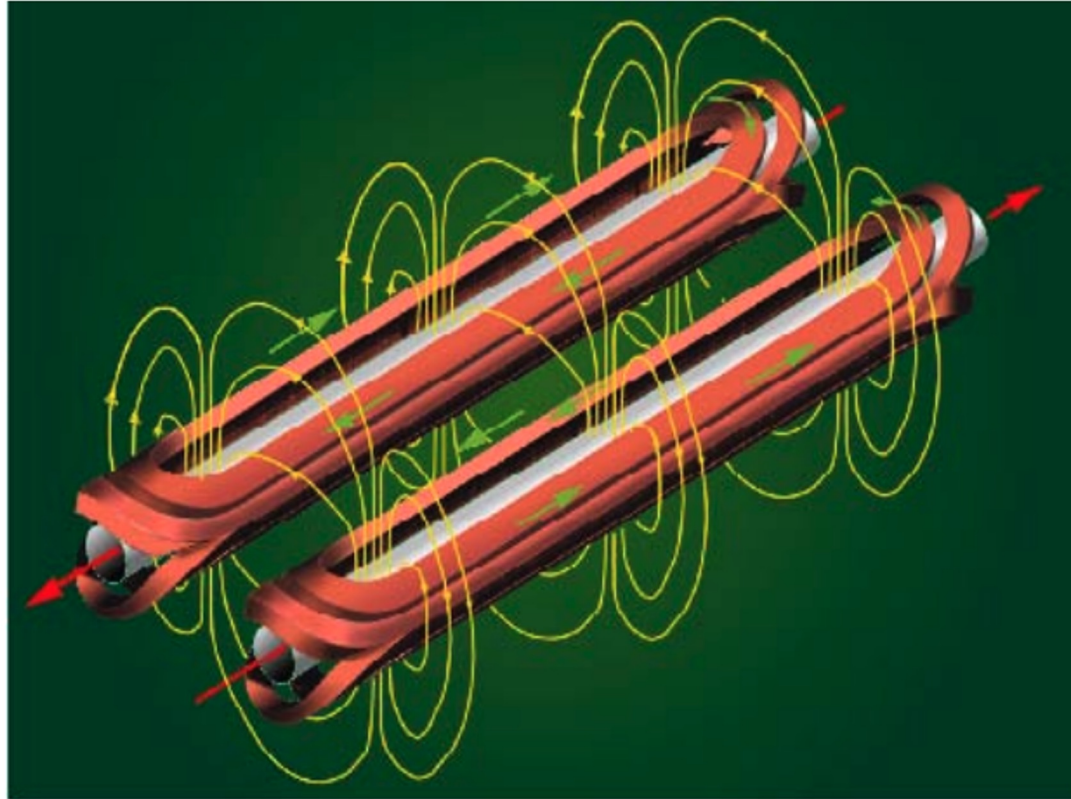
Dipole edge effects

Recall from our betatron in Lecture 2: gradient at the edge, used to give vertical focusing (weak focusing)



Dipole conformed to beam pipe

Dipole Field Produced by SC Tape



Some quadrupole configurations

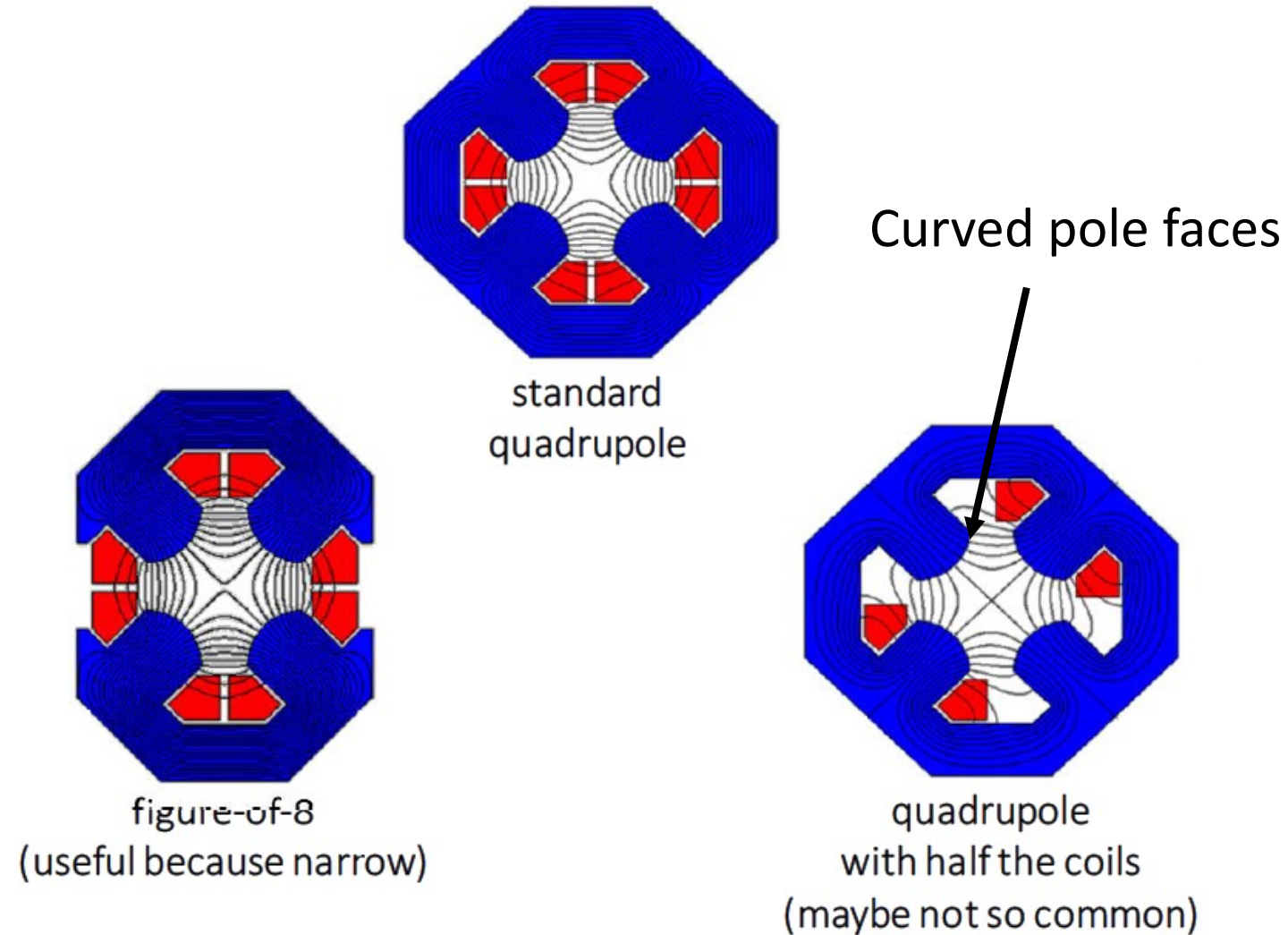


Figure: https://indico.cern.ch/event/1101643/contributions/4635257/attachments/2372240/4058445/JAI_course_Jan_2022_handouts.pdf

Sextupole example

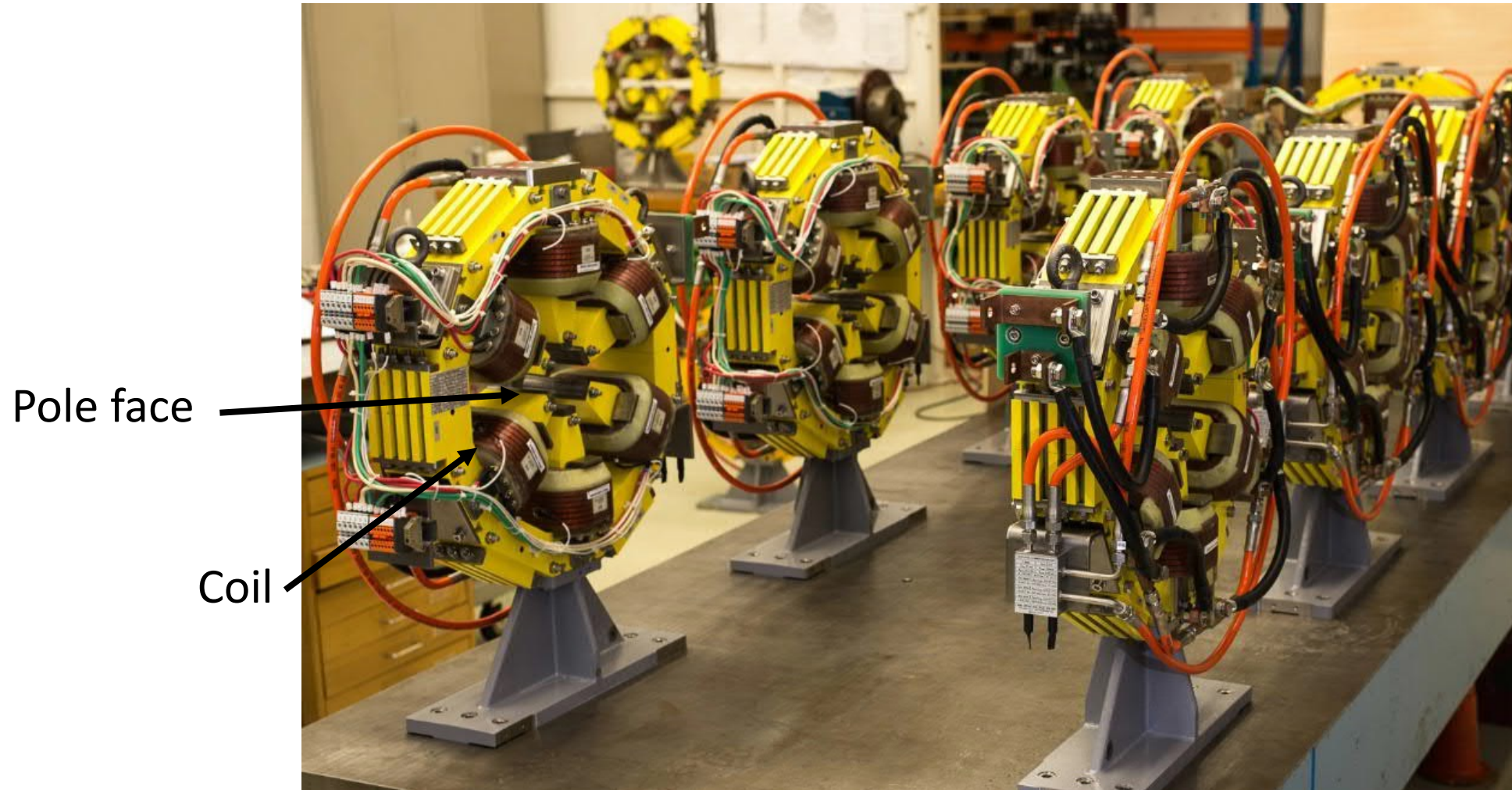


Figure: https://indico.cern.ch/event/1101643/contributions/4635257/attachments/2372240/4058445/JAI_course_Jan_2022_handouts.pdf

Design choices

- Coils made from some conductor or superconductor
 - Superconducting case: needs to be cooled to cryogenic temperatures for superconductivity
 - Resistive case: Ohmic losses cause heating, requires some water cooling
- Coil design: material choice, wire gauge (physical thickness), number of turns, density of turns
- Pole design: pole faces can be shaped to provide optimized fields to counteract errors, edge effects, etc.
- How:
 - Magnets in accelerator: lattice codes
 - Physical magnets: finite element modeling of both magnetic field and mechanical factors (heating, stress)

Finite element modeling

- Geometry broken up into mesh of elements
- Equations of the system solved over those elements
- Boundary conditions matched iteratively
- Broadly applicable: E&M, thermodynamics, fluid dynamics
- Also how we model cavities: see more on Monday
- Result: can model the shape of the field produced in the beam pipe

